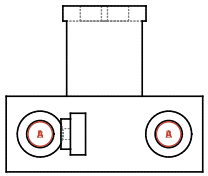
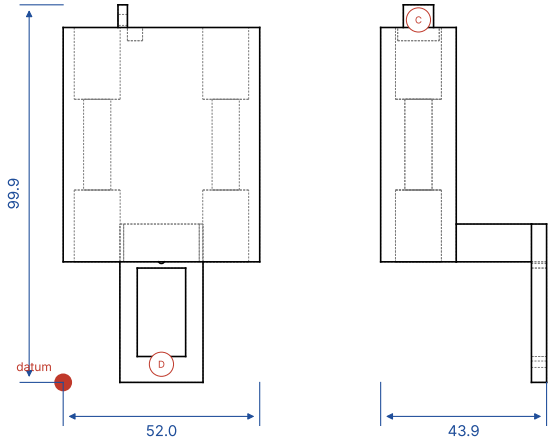
CA-301 Carriage + arm

NOTES

1. Hole sizes are the printed sizes from the PETG tolerance table (backend/cad/model.py TOL). Print CA-900 first and adjust TOL if a fit is off.
2. Screws thread straight into the printed pilots (HF kit #67675, no nuts).
3. Datum = the part's bounding-box corner (min X, Y, Z), lower left of the front and top views. Hole centres: (X, Y) for Z holes, (X, Z) for Y holes, (Y, Z) for X holes. ×k = k holes stacked along the axis; +n = n more positions (see the STEP file).
4. Shown in its assembly orientation; the STEP file defines all other geometry.

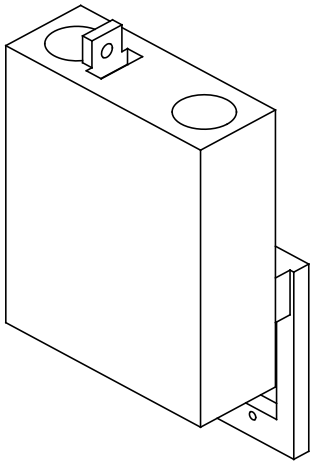
TOP VIEW





FRONT VIEW

RIGHT VIEW



ISOMETRIC (not to scale)

HOLE TABLE (letters mark hole centres on the view along that axis)

	Ø (mm)	×	axis	depth	centres from datum (mm)	fit / feature
A	12.15	4	Z	19.0	(9.0, 10.0) ×2; (43.0, 10.0) ×2	LM6UU press fit
B	7.20	2	Z	24.0	(9.0, 10.0); (43.0, 10.0)	#6 pan-head pocket
C	2.90	1	X	2.5	(10.0, 95.9)	#6-32 pilot (self-threading)
D	1.70	1	Y	4.0	(26.0, 4.8)	servo-bag screw pilot

PART NO.	**CA-301**
TITLE	**Carriage + arm**
MATERIAL	PETG printed, 0.4 mm nozzle
MASS	45.8 g (CAD, 60 % fill)
QTY	1 per robot
SCALE	1:2
UNITS	mm · general tolerance ±0.2
SOURCE	cad/step/parts/carriage_arm.step (from backend/cad/model.py)
PROJECTION	Third angle ⊕◁
SHEET	7 of 13
DRAWN	ME 2331 Project 2 team 2026-11-02